Make

Make is a build automation tool. It builds executable programs and libraries from source code by following instructions in *Makefiles* to derive the target program. As IDEs and language-specific compilers find prominent use on Windows and other platforms, *make* is widely used on UNIX (-like) systems. *Make* also decides whether the target needs to be regenerated or not, by comparing file modification times and thus avoiding rebuilding files that are already up-to-date.

Git

Initially designed by Linus Torvalds for the Linux kernel source, Git is a distributed revision control and source-code management system with an emphasis on speed. Each Git working directory is a full-fledged repository with complete revision-tracking capabilities, which does not need network access or a central server for history and other functions. Git's main features include distributed development, compatibility with existing systems and protocols, efficient handling of large products, a tool-kit based design, and periodic explicit object packing.

Installation

Now let's move on to installation.

Xcode on Lion and Mountain Lion

Visit *https://developer.apple.com/xcode/* and go to *'View in Mac App Store'*. This will automatically redirect you to the Xcode page of the App Store on your Mac. Just click *Free*, then *Install App*.

Xcode on Snow Leopard

On Leopard, it's preferable to install Xcode 4.2. Go to *http://developer.apple.com/downloads/* and sign in with your Apple ID. If you're not registered yet, do it now. Search for Xcode 4.2, click *'Xcode 4.2 for Snow Leopard'* and then click on the *.dmg* link to download it. Once the download is finished, it will automatically launch a window in your Finder, as shown in Figure 1. Double-click the Xcode package installer to launch it. See that all checkboxes are selected. Wait till the installation completes.

Installing CLT

Once the Xcode installation is complete, you can install any further components from Xcode. Launching Xcode from the *Applications* folder and from the menu bar go to *Xcode > Preferences*. Click the *Install* button for *'Command Line Tools'* to install it (see Figure 2).

Installing Git

Installing Git is simple; visit *http://git-scm.com/downloads/;* click *'Mac OS X'*, and your *.dmg* is downloaded. From your Finder, launch it to install Git on your system. After you've

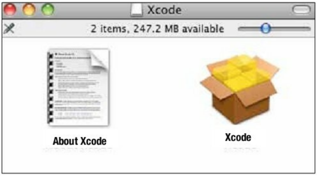

Figure 1: The .dmg file auto-opened

Figure 2: Install CLT in Xcode

installed Git, configure it! Launch the Git Bash shell and run the following commands:

```
$ git config --global user.name "your name"
$ git config --global user.email "youremail@email.com"
```

These commands set the name and email address for Git to use when you commit. Git commit is a utility which helps record a snapshot of staging content.

Usage

The CLT can be used from Xcode, and also independently from the terminal. For usage and parameters, please do check the documentation for these tools, which is available online. Happy developing! END

References

[1] *http://en.wikipedia.org*
[2] *http://www.moncefbelyamani.com*

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